

Assure the campaign, not just the transaction.

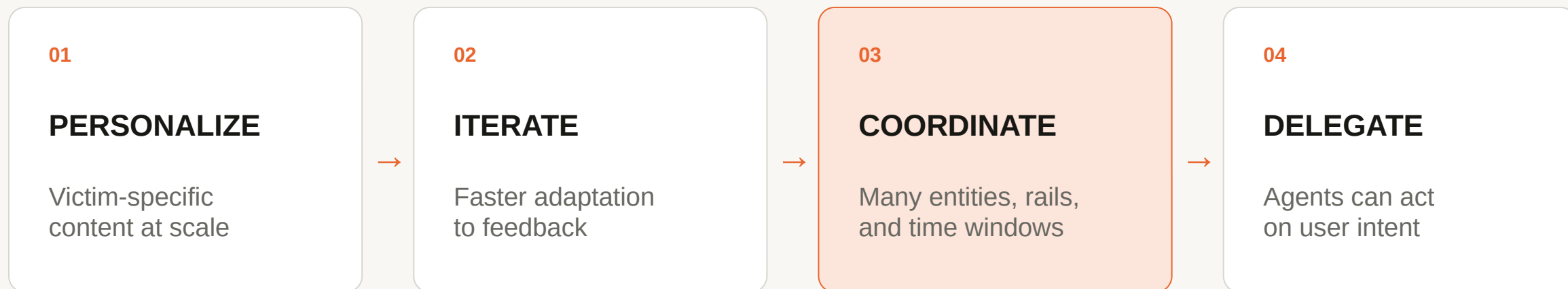
APAR turns emerging GenAI payment threats into rail-correct synthetic campaigns, tests layered defenses, and preserves a human promotion boundary.

SYNTHETIC • OFFLINE • REPLAYABLE

Dylan Moraes • Competition submission



GenAI changes the operating model of fraud



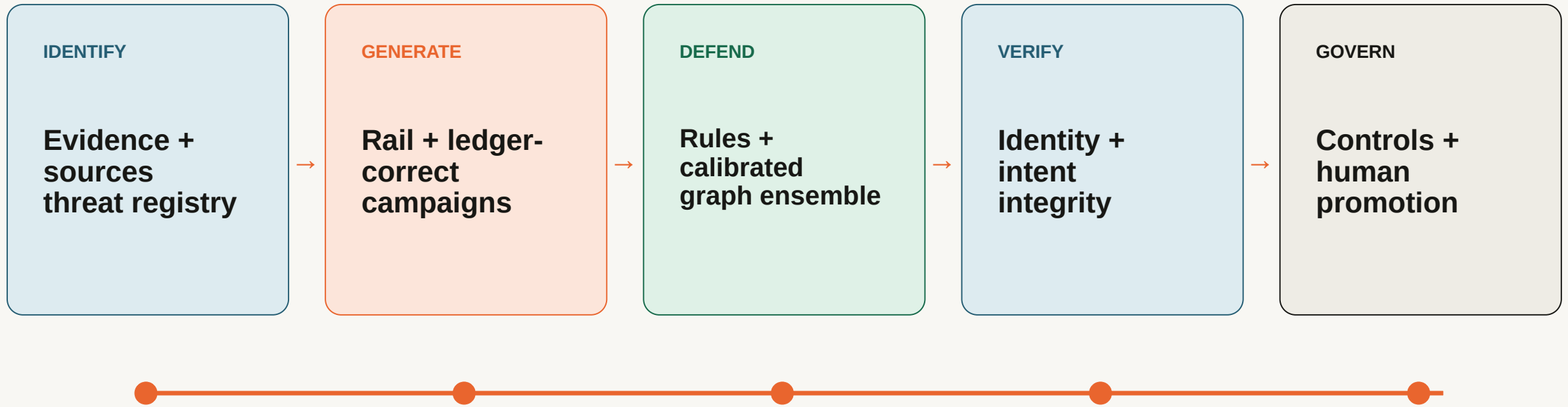
Transaction classifier



Campaign-aware assurance system

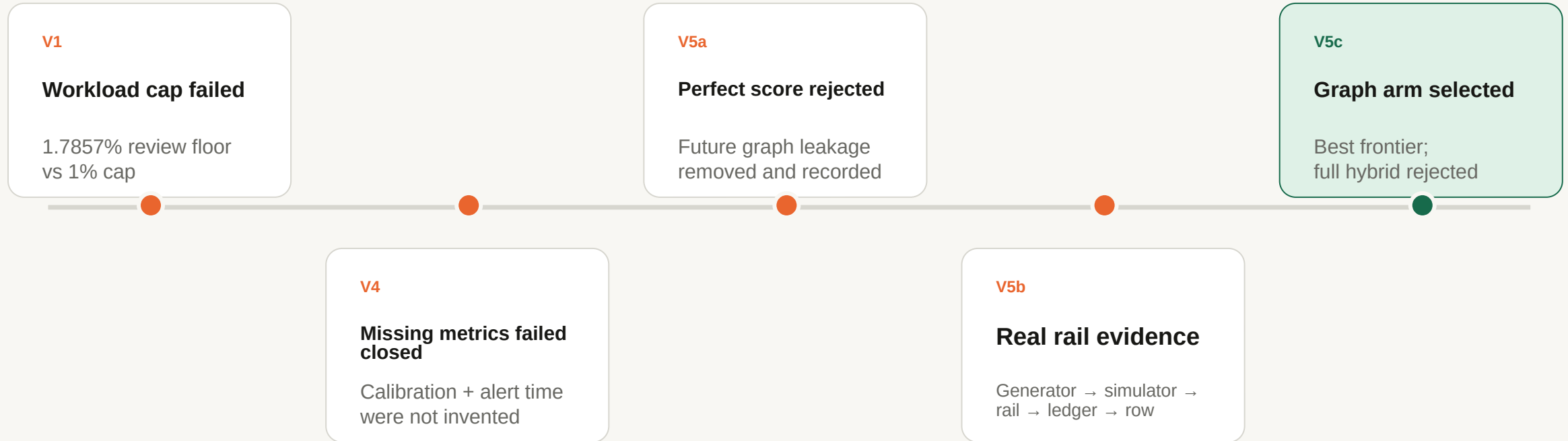
Behavior + lifecycle + economics + authority + governance

One assurance loop connects five control planes



Bounded feedback returns failed controls to the research backlog—never directly to deployment.

Failed experiments shaped the winning architecture



Novelty: not a single algorithm—an evidence-governed loop that learns from its own failure modes.

Graph context wins the operational frontier

Verified recovered diagnostics — synthetic and non-authoritative

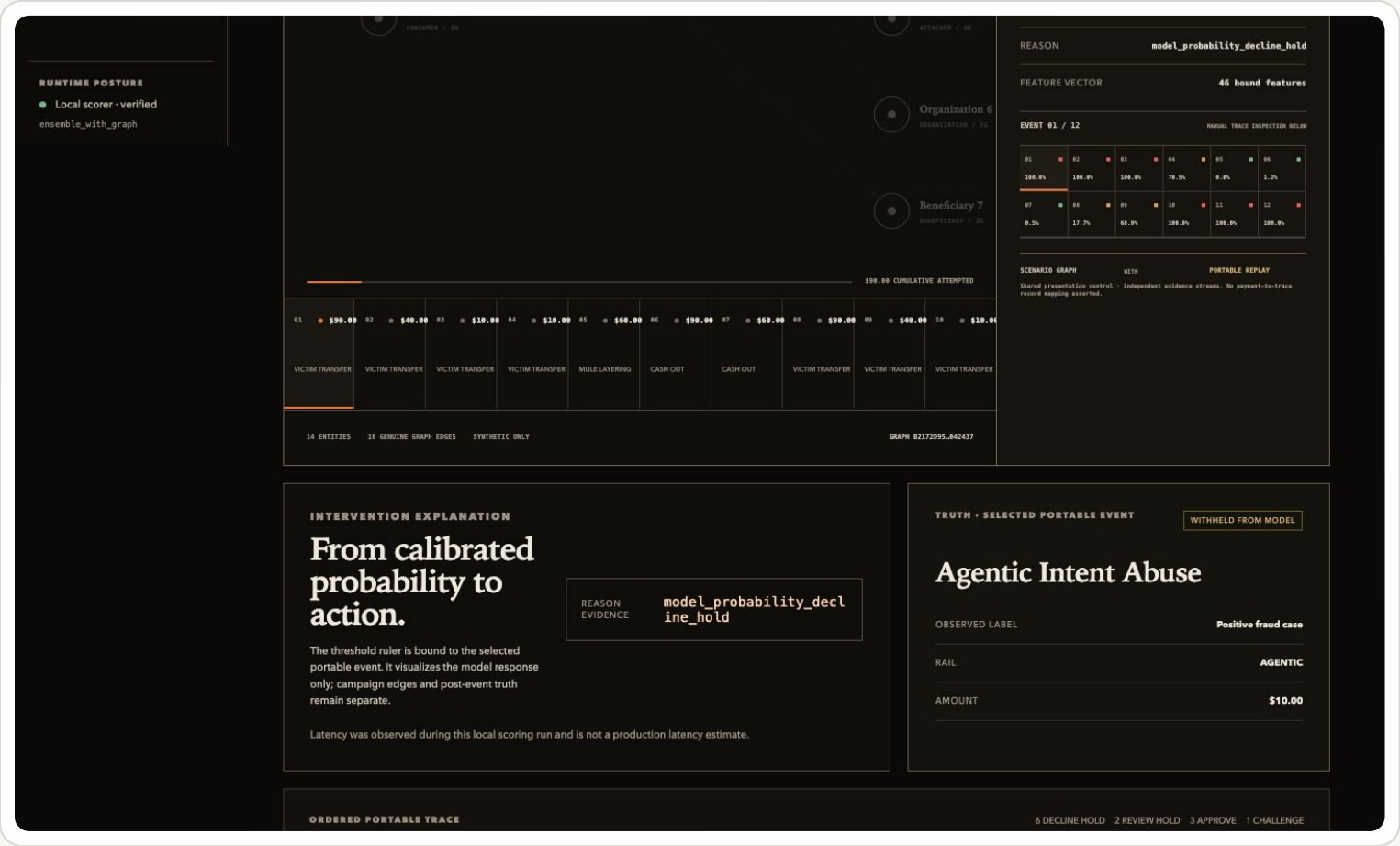
ARM	RECALL %	PRECISION %	F1 %	FALSE-DECLINE %	P95 MS
Rules only	85.95	14.88	25.37	87.44	4.06
Ensemble • no graph	99.74	94.76	97.19	0.000	3.38
Ensemble • graph	99.87	95.88	97.83	0.0037	3.54
Full hybrid	99.93	16.85	28.83	87.44	19.74

Precision +1.12 pts

Challenge -0.23 pts

Latency +0.16 ms p95

A small, portable model powers the live demo



3 × CatBoost members

46 frozen causal features

12 hash-bound scenarios

0.0 max replay error

Approve • Challenge • Review hold • Decline hold

Investigators see one campaign—not twelve alerts

CASE-CENTRIC

14

entities

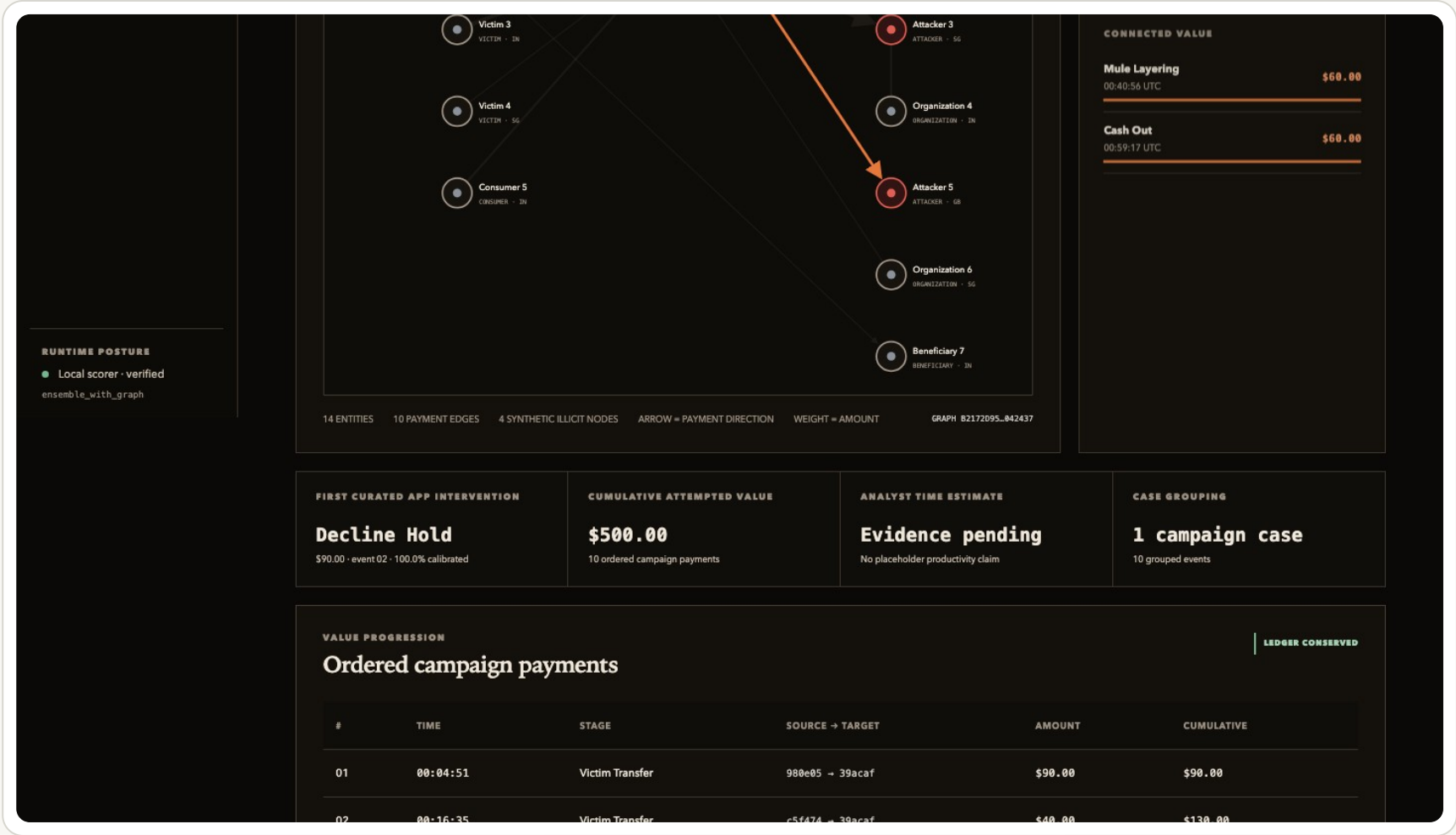
10

payment edges

\$500

conserved synthetic ledger

Analyst-time benefit: evidence pending



Agentic commerce needs proof before probability



Agent identity



Merchant + cart



User mandate



Expiry + nonce



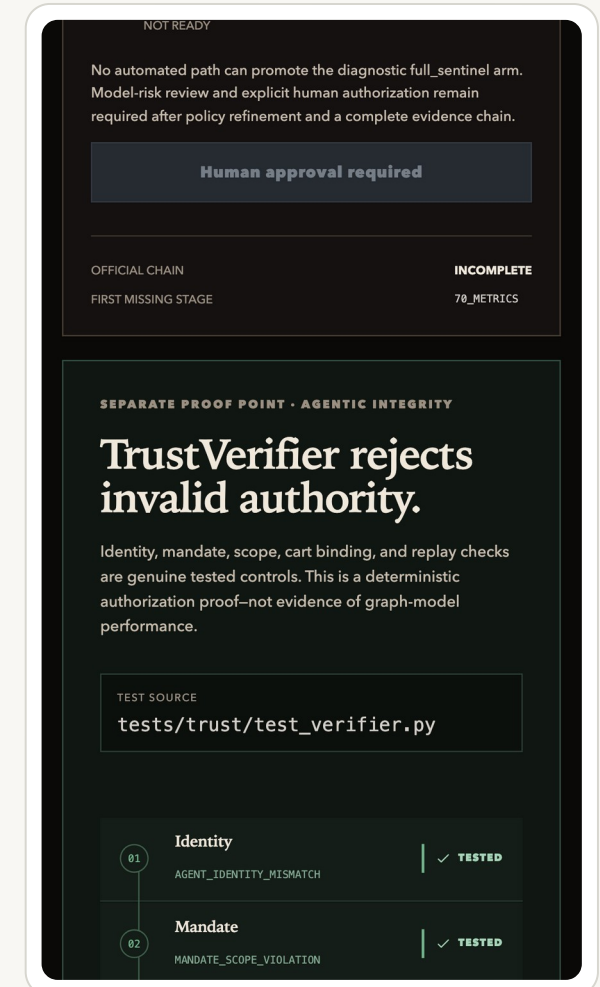
Scope + amount



Replay rejection

DEFINITIVE FAILURE

DECLINE_HOLD before statistical risk



Evidence governance is part of the product

01

Causal features

Future append +
equal-time isolation

02

Executed controls

Shuffle, rename,
leakage, single-class

03

Immutable evidence

Models, traces,
metrics, manifests
hashed

04

Independent replay

Fresh environment re-
scores all 12 cases

05

Human promotion

The model cannot
approve itself

Portable bundle

52ed4c31...dbc900

Official chain

Stops at Stage 60

STAGE 70 NOT CLAIMED

A sharp claim boundary makes the demo credible

WHAT WE PROVE

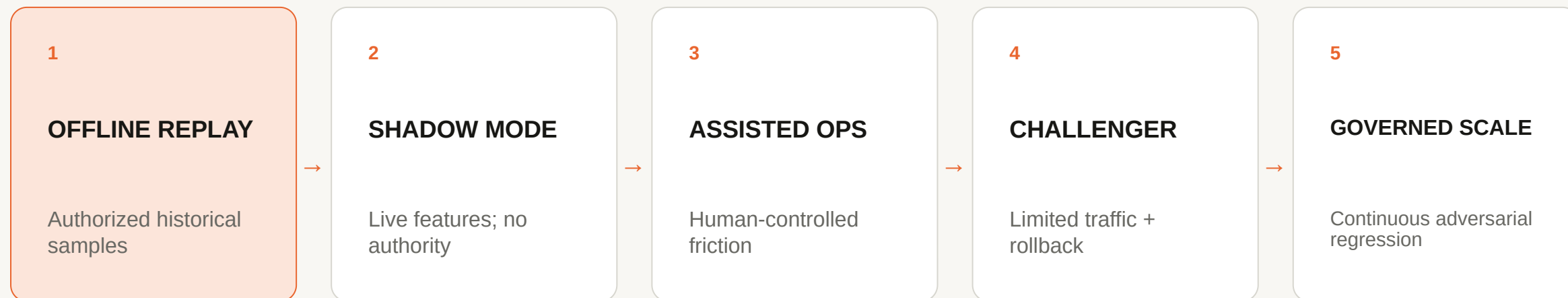
- ✓ Portable model loads and replays exactly
- ✓ Four rail-backed synthetic families
- ✓ Graph/no-graph architecture comparison
- ✓ Deterministic TrustVerifier integrity
- ✓ Offline console + hash-bound fallback

WHAT WE DO NOT CLAIM

- ✗ Production or real-cardholder performance
- ✗ Accepted official Stage 70 capacity evidence
- ✗ External or multi-institution validation
- ✗ Autonomous model promotion
- ✗ Full Sentinel as the champion

Synthetic evidence before assertion.

From competition prototype to governed payment assurance

**APAR**

**Assure the campaign. Verify the intent.
Promote only the evidence.**

DEMO READY